Extended Abstract Track

Rank- and Scale-Conditioned Curvature Is Associated with Local Linear Decodability in Vision Representations

Editors: NeurReps 2026 organisers

Abstract

Geometric summaries of neural activations depend on the rank and neighbourhood scale of the local chart used to compute them. We evaluate nested regularised quadratic charts of a ViT-B galaxy representation, define sphere-normal mean curvature K_H^{cross} from a split-half second fundamental form, and select chart ranks from held-out reconstruction rather than a single eigengap. At the frozen neighbourhood k=2048 (n=512 anchors), greater curvature is associated with poorer local out-of-fold linear decodability of r-band magnitude (`mag_r_desi_local_oof_r2`) at the middle and upper evaluated ranks: controlled Spearman $\rho = -0.240$ at d=16 and $\rho = -0.233$ at d=20. Direct-error associations have the opposite sign (MAE $\rho = +0.251$, MSE $\rho = +0.227$), while local SST/target variance is essentially unassociated ($\rho \approx -0.025$). On n=128 hash-selected anchors the middle/upper direction remains negative at k=1024 and k=1536, but magnitudes attenuate substantially; k=512 fails curvature reliability and is not confirmatory. The association is therefore supported at the frozen scale and scale-dependent. Scope: one encoder, one astronomical dataset, one probe target, an exploratory rank sweep, and no independent replication.

Keywords: representation geometry, mean curvature, linear probes, vision transformers, chart rank

1. Question and contributions

Shared geometric structure is a working hypothesis for foundation-model activations (Huh et al., 2024; UniverseTBD et al., 2025). We ask a narrower question about one encoder: after a nested quadratic chart is fit in a finite neighbourhood, is its sphere-normal mean curvature associated with how well a *fixed* global linear probe (Alain and Bengio, 2017) reconstructs a physical scalar on that neighbourhood?

The probe is five-fold ridge regression from a ViT-B (Dosovitskiy et al., 2021) embedding to r-band magnitude. The outcome at an anchor is the local out-of-fold coefficient of determination

$$R^2_{\mathrm{local}} = 1 - \mathrm{SSE}_{\mathrm{OOF}}/\mathrm{SST}_{\mathrm{local}},$$

denoted `mag_r_desi_local_oof_r2`. Catalog magnitude `mag_r_desi_catalog_value` is a different vector and is never substituted. Curvature K_H^{cross} is the split-half inner product of mean-curvature vectors of a sphere-normal second fundamental form: finite-scale, extrinsic, and conditioned on chart rank and neighbourhood size.

Contributions. (i) A method for evaluating local representation dimensionality as a predictive range rather than selecting one eigengap. (ii) A sphere-normal quadratic-chart curvature statistic that removes the forced radial curvature induced by activation normalisation. (iii) A controlled empirical association between rank-conditioned curvature and local OOF probe error at a frozen neighbourhood scale. (iv) An audit showing that the association is not an R^2-denominator artefact and is materially neighbourhood-scale dependent.

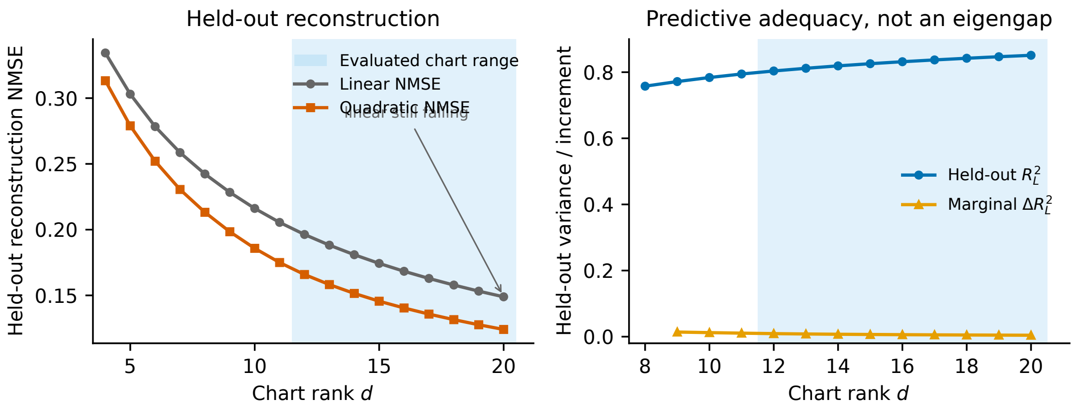

Figure 1: Chart rank is evaluated by held-out reconstruction, not a single eigengap. Left: pooled linear and regularised quadratic NMSE versus chart rank (k=2048). Right: pooled R_L^2 and its increment. Shading marks the *evaluated chart range* d=12–20 ($\tau \approx 0.80$ and 0.85). Linear NMSE is still falling at d=20; no vertical line is drawn at a putative intrinsic dimension.

2. Local quadratic geometry and chart rank

At each of 512 hash-stable anchors we take the k=2048 nearest neighbours on the unit sphere, remove the radial component, and compute a nested PCA frame J, following the local-quadratic charting tradition (Donoho and Grimes, 2003). A two-stage ridge quadratic is fit with split-half cross-evaluation: a tangential warp, then a second fundamental form B^S projected onto the sphere-normal complement of span(x_0, J). Packed off-diagonal coefficients store $2B_{ab}$. Mean curvature is $H_a = d^{-1} \sum_i B^S_{a,ii}$; the production scalar is $K_H^{\text{cross}} = \langle H^{(A)}, H^{(B)} \rangle$. Reliability R_H is the split-half concordance of H. Identities appear in the appendix.

Held-out reconstruction (Fig. 1) does not exhibit a single spectral jump. Pooled linear R_L^2 is approximately 0.80 at d=12 and 0.85 at d=20; quadratic NMSE decreases more slowly at higher ranks, while *linear* reconstruction continues to improve at the right edge of the plotted family. We therefore treat d=12–20 as the *evaluated chart range* spanning roughly 80–85% explained variance, not as a complete geometrically adequate dimensional interval and not as an intrinsic dimension. A stable lower-dimensional core is compatible with further predictive gains at higher ranks. Chart positions used for scale mapping were fixed geometrically before inspecting curvature–probe correlations: lower d=12, middle d=16, upper d=20. Rank scanning inside the family is exploratory.

3. Curvature and local decodability

Primary frozen analysis. Aligning K_H^{cross} and `mag_r_desi_local_oof_r2` by `sample_id` on n=512 anchors at k=2048 reproduces the audited controlled Spearman correlations: $\rho_{12} = +0.143$, $\rho_{16} = -0.240$, $\rho_{20} = -0.233$ ($\rho_{20} - \rho_{12} = -0.376$). The corresponding raw

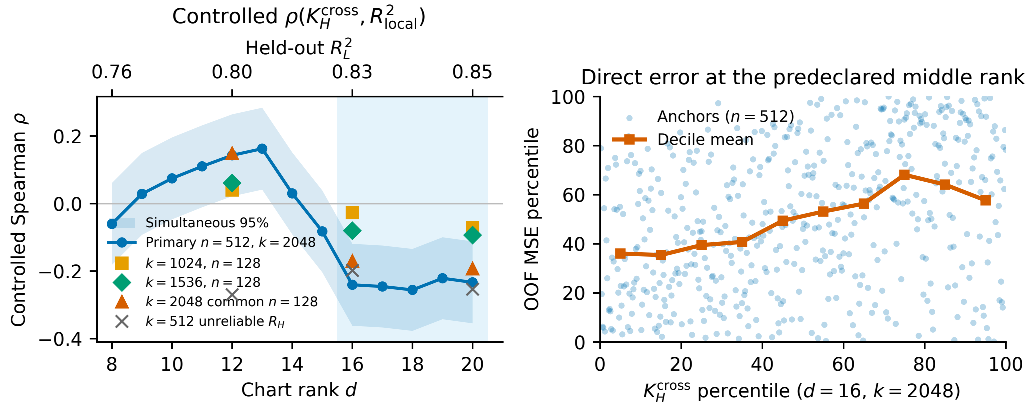

Figure 2: Finite-scale, rank-conditioned association between curvature and local probe error. Left: controlled $\rho(K_H^{\text{cross}}, R^2_{\text{local}})$ versus chart rank for the primary n=512, k=2048 analysis (line, simultaneous 95% band). Markers show n=128 scale estimates; k=512 is marked unreliable. The top axis is held-out R^2_L. Shading: FWER-negative span d=16–20 in the primary analysis. Right: percentile relationship between K_H^{cross} and OOF MSE at the predeclared representative rank d=16 (primary n=512, k=2048). Catalog magnitude and DESI labels are not shown.

(uncontrolled) correlation at d=16 is $\rho = -0.412$. The reconstructed R^2_{local} matches the frozen local-probe table to numerical identity. A typed schema rejects catalog magnitude as the primary target.

Controls are log kNN radius, local label variance, and evaluation count. Permutations (B=10^4) use rank-space Freedman–Lane for the controlled family; p=0 is reported as p<$1/(B+1)$. Familywise error is taken over ranks d=$12, \ldots, 20$. The negative controlled associations at d=16 and d=20 have FWER p<10^{-4}. Paired bootstrap (B=2000) simultaneous 95% intervals: $\rho \in [-0.323, -0.156]$ at d=16 and $\rho \in [-0.317, -0.147]$ at d=20. At d=12 the controlled association is positive ($\rho = +0.143$, FWER p=0.013). The rank that maximises $|\rho|$ is not treated as an intrinsic-dimension estimate.

Direct error versus denominator. Greater curvature corresponds to *higher* local prediction error and therefore *lower* R^2_{local}. After the same controls, at d=16, k=2048, K_H^{cross} is associated with OOF MAE ($\rho = +0.251$, 95% CI $[0.154, 0.333]$) and OOF MSE/SSE ($\rho = +0.227$, $[0.123, 0.317]$), both FWER p<10^{-4}, and is essentially unassociated with SST / local target variance ($\rho \approx -0.025$, FWER p=1.00). NMSE equals $1 - R^2_{\text{local}}$ and is not an independent metric. Evaluating probe metrics on the outer neighbour half (indices $[k/2, k)$) while leaving K_H^{cross} unchanged yields $\rho_{16} = -0.246$ for R^2_{local} and +0.238 for MSE. A label shuffle at d=16 gives controlled $\rho \approx -0.059$.

Scale sensitivity. The n=512, k=2048 result is the frozen primary analysis. Scale comparisons use n=128 hash-selected anchors and are not power-matched to the primary test.

At the predeclared middle/upper ranks the direction remains negative at k=1024 and k=1536, but magnitudes attenuate substantially (controlled $\rho_{16} = -0.027$ and -0.080). The k=2048 common-128 estimate at d=16 is $\rho = -0.171$. Neighbourhood k=512 fails split-half curvature reliability (median $R_H \approx -0.19$) and is not interpreted as evidence of a contradictory scientific association. We do not apply post-hoc attenuation correction. Decision label: `claim_supported_but_scale_dependent`.

4. Limitations and conclusion

K_H^{cross} is finite-scale extrinsic curvature of a fitted local chart of the activation cloud, not the intrinsic Riemannian curvature of an unknown manifold, and it is defined only at a chosen chart rank and neighbourhood size. The evaluated range d=12–20 does not identify a unique intrinsic dimension. The association is neighbourhood-scale sensitive. We study one encoder, one dataset, and one probe target. The rank sweep inside d=8–20 was not pre-registered. There is no independent encoder or dataset replication. No causal interpretation is offered. DESI spectroscopic labels are excluded because sample-identity alignment was not established. Catalogue-magnitude correlations are not reported as discovery results.

Conclusion. At the frozen k=2048 scale, rank-conditioned sphere-normal mean curvature is associated with worse local OOF linear-probe performance at the middle and upper evaluated chart positions. The association survives direct-error and denominator checks, but its magnitude attenuates substantially at intermediate neighbourhood scales.

References

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2017. ICLR 2017 Workshop.

David L. Donoho and Carrie Grimes. Hessian eigenmaps: Locally linear embedding techniques for high-dimensional data. *Proceedings of the National Academy of Sciences*, 100 (10):5591–5596, 2003. doi: 10.1073/pnas.1031596100.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The Platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20617–20642. PMLR, 2024. URL https://proceedings.mlr.press/v235/huh24a.html.

UniverseTBD, Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The Platonic universe: Do foundation models see the same sky? *arXiv preprint arXiv:2509.19453*, 2025. doi: 10.48550/arXiv.2509.19453.

Appendix A. Implementation notes

Definitions are taken from the code paths that produced the frozen artifacts; see also `METHODS_FOR_PAPER.md`.

Local coordinates: row-ℓ_2 unit embeddings; x_0 normalised; radial component removed before nested PCA; $J \leftarrow \mathrm{orth}((I - x_0 x_0^\top)J)$ (`sphere_project_basis`). Quadratic features $\phi_{ab} = u_a u_b$ for $a \leq b$; off-diagonal packed entries store $2B_{ab}$. B^S columns are projected onto the complement of $\mathrm{span}(x_0, J)$. Production K_H^{cross} is the split-half inner product of H, not $\|H\|$. OOF probes are five-fold ridge (α=100). Local R^2 uses uniform weights on finite neighbours:

$$R^2_{\mathrm{local}} = 1 - \sum_i (y_i - \hat{y}_i)^2 \Big/ \sum_i (y_i - \bar{y})^2.$$

SSE is $\sum_i (y_i - \hat{y}_i)^2$; MSE is SSE/$n_{\mathrm{eval}}$; MAE is $n_{\mathrm{eval}}^{-1} \sum_i |y_i - \hat{y}_i|$; SST is $\sum_i (y_i - \bar{y})^2$. NMSE is $1 - R^2_{\mathrm{local}}$ and is therefore not an independent estimand.

Appendix B. Target identity and parity

The primary target is local OOF probe R^2 (`mag_r_desi_local_oof_r2`). Catalog magnitude (`mag_r_desi_catalog_value`) is a distinct vector. A typed schema rejects ambiguous labels (`local_r2`, `mag_r_desi`) and rejects a catalog vector that is identical or an affine rescaling of the probe-performance target. After `sample_id` alignment, controlled Spearman correlations at k=2048, n=512 match the frozen audit to the reported rounding: $\rho_{12} = +0.143$, $\rho_{16} = -0.240$, $\rho_{20} = -0.233$. Reconstructed R^2_{local} matches the frozen geography table with maximum absolute difference 0.

Appendix C. Probe-metric associations at k=2048

Table 1 reports confirmatory controlled Spearman associations at the predeclared ranks (primary n=512). SST/variance remain near zero after controls; MAE/MSE reverse the R^2 sign, as required of an error association. Permutation p=0 is written $p<10^{-4}$ (B=10^4). Intervals are paired-bootstrap 95% CIs; FWER is over d=12, …, 20.

Table 1: Controlled $\rho(K_H^{\mathrm{cross}}, \cdot)$ on n=512 anchors, k=2048.

Target at d=16	ρ	95% CI	FWER p
`mag_r_desi_local_oof_r2`	−0.240	[−0.323, −0.156]	$<10^{-4}$
OOF MAE	+0.251	[0.154, 0.333]	$<10^{-4}$
OOF MSE / SSE	+0.227	[0.123, 0.317]	$<10^{-4}$
Local SST / target var.	−0.025	—	1.00
NMSE = $1 - R^2_{\mathrm{local}}$	+0.240	—	—
R^2_{local} at d=12	+0.143	[0.051, 0.226]	0.013
R^2_{local} at d=20	−0.233	[−0.317, −0.147]	$<10^{-4}$

Raw (uncontrolled) Spearman at d=16 for R^2_{local} is $\rho = -0.412$. Outer-half neighbour probe metrics (indices $[k/2, k)$, K_H^{cross} unchanged) at d=16: R^2_{local} $\rho = -0.246$, MSE $\rho = +0.238$. Label shuffle at d=16: controlled $\rho \approx -0.059$.

Appendix D. Scale subset and reliability

Chart positions were fixed from held-out variance (τ=0.80, 0.85 and the midpoint), not by maximising $|\rho|$. Table 2 uses the rank-sweep 128-anchor refits. Signs at d=16, 20 stay negative at k=1024 and 1536, but magnitudes are <50% of the primary k=2048, n=512 value. Split-half reliability R_H fails at k=512 (median ≈ -0.19) and recovers to ≈ 0.24 at k=1024 and ≈ 0.41 at k=1536. The k=512 row is reported for completeness and is not confirmatory.

Table 2: Controlled $\rho(K_H^{\text{cross}}, R^2_{\text{local}})$ at predeclared ranks. Primary k=2048 uses n=512; other rows use n=128.

k	d=12	d=16	d=20
2048 (n=512)	+0.143	−0.240	−0.233
512 (unreliable R_H)	−0.270	−0.198	−0.253
1024	+0.040	−0.027	−0.072
1536	+0.061	−0.080	−0.093
2048 common 128	+0.148	−0.171	−0.193

Appendix E. Excluded analyses

DESI spectroscopic labels are omitted because embedding-table identity alignment is unproven; equal row counts are not a join. Catalog-magnitude associations are retained only as a negative control against silent substitution and are not a discovery result. Adaptive-run physics labels other than the frozen probe target are not used for the claim. An earlier catalog-label substitution motivated strict target typing; it is not part of the scientific narrative of the main text.